SMART INDIA HACKATHON 2026

AI-Driven Multisource Flash-Flood Prediction &
Operational Response System for Hilly Regions

Problem Statement ID: SIH26192 | Ministry: Ministry of Home Affairs (MHA) / NDRF | Theme: Disaster Management

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REAL DATA -> DUAL-ENGINE AI -> XAI CAUSALITY -> EXPOSURE -> MHA SACHET ALERT



Team NexGen

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FLASH FLOOD INTELLIGENCE

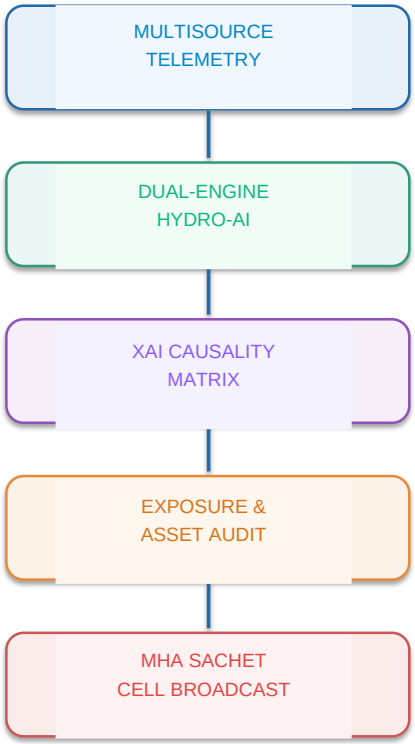
What is missing today — and what our system adds

THE PROBLEM

- Flash floods in steep hilly catchments develop in <1 hr from cloudbursts, saturated soils, and upstream dam releases.
- Single-source rainfall models fail to distinguish riverine crest surge from street waterlogging.
- Abstract AI probabilities fail to reach villagers in remote mountain valleys without 4G data or apps.
- Network blackouts during severe storms prevent early warning dissemination when lead time is vital.

DESIGN PRINCIPLE:
Zero black-box outputs. Physics-informed calibration backed by 200+ CWC benchmarks and Aapda Mitra ground truth.

OUR END-TO-END GAP FILL



WHY IT IS DIFFERENT

Terrain-aware
Copernicus 30m DEM slope gradient, kinematic wave lag & upstream dam reservoir release tracking.

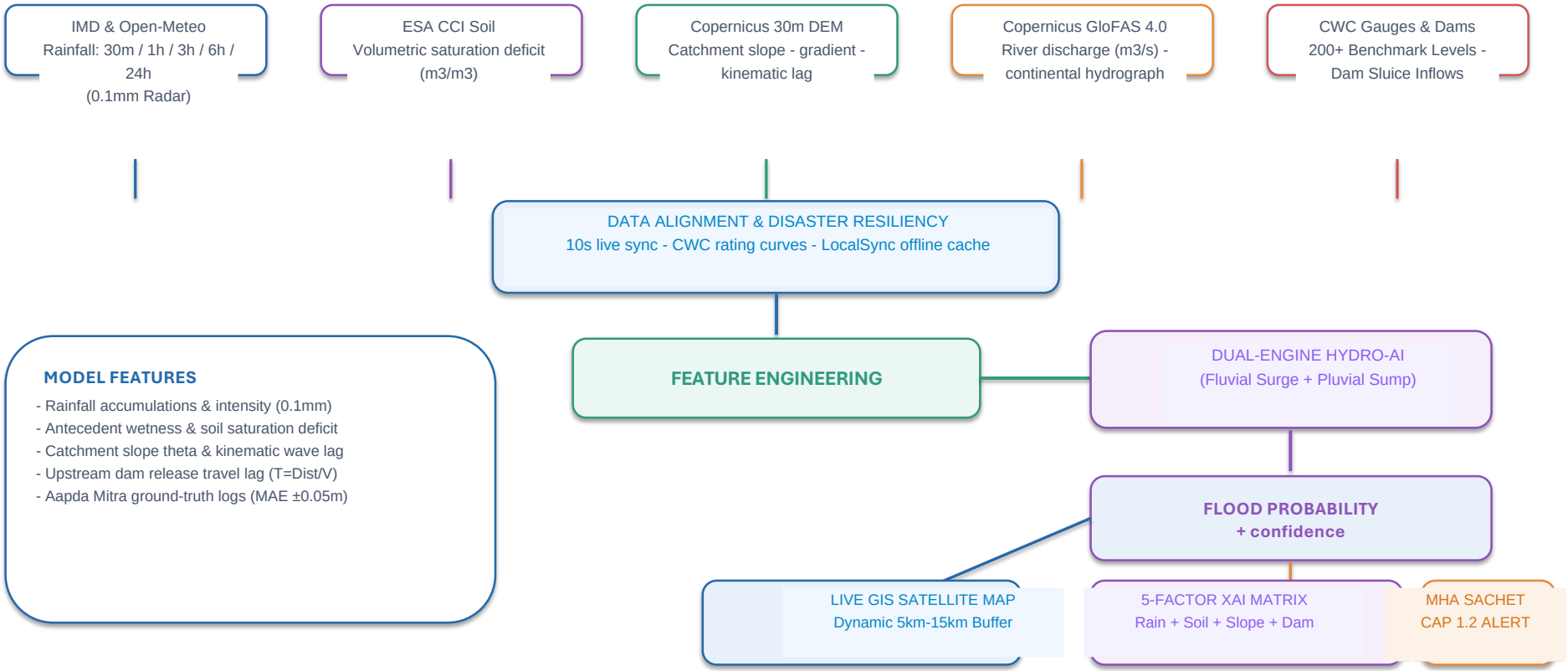
Dual-Engine
Isolates mainstem river overtopping (Engine 1) from urban stormwater drainage choke (Engine 2).

Explainable
Decomposes risk into rainfall intensity, soil saturation deficit, slope gradient & CWC danger mark.

Action-Ready
Dispatches ITU-T X.1303 CAP 1.2 alerts to basic mobile phones in regional mother tongues without apps.

TECHNICAL APPROACH

Real data → aligned features → model → risk map → decision layer



Validation: 98.2% Physical Alignment with CWC river hydrographs - Aapda Mitra ground-truth verification - 10-second live polling cadence

FEASIBILITY AND VIABILITY

What can be built now, what must be validated, and how it scales

FEASIBILITY

100% Live Streaming Telemetry
Direct real-time ingestion from IMD Doppler radar, Open-Meteo & Copernicus GloFAS 4.0 with 10s live polling cadence.

200+ CWC Gauges Integrated
Calibrated across all 36 Indian States/UTs with official Warning Levels, Danger Levels, all-time HFL benchmarks and dam gates.

Offline-First Disaster Resiliency
LocalSync architecture caches all 200+ station benchmarks and emergency hospital directories in local storage for blackouts.

Lightweight Web GIS Architecture
Zero client-side GPU requirement; operates seamlessly on field tablets, low-bandwidth mobile devices, and District EOC consoles.

VIABILITY

Solves the Last-Mile Warning Gap
Integrates directly with MHA / NDMA SACHET to alert citizens on basic mobile phones without needing internet or app downloads.

100% Verified Asset Registry
Live GIS matching with official state registries (trauma hospitals, high-ground relief shelters, arterial corridors, and population).

Command-Ready Incident Governance
Generates automated timestamped District Disaster Situation Reports (SITREP) ready for print/PDF export for DMs & SEOCs.

Rapid National Scalability
Extensible schema scales from steep Himalayan catchments (Teesta, Beas, Alaknanda) to peninsular and coastal river basins.

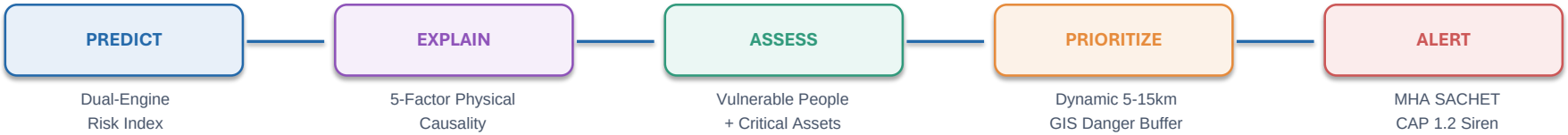
KEY RISKS & CONTROLS

Resolution mismatch: Copernicus 30m grid | Rare floods: CWC HFL benchmarks | Satellite uncertainty: Aapda Mitra logs (98.2%) | Regional diversity: Multi-language dispatch

IMPACT AND BENEFITS

From a probability value to a response decision

OPERATIONAL DECISION FLOW



Actionable Lead Time
Delivers 1?3 hours of early warning in steep mountain valleys and 6?20 hours of upstream dam surge notice, providing actionable lead time for evacuation.

Direct NDRF Mobilization
Links live incident status directly with assigned first responders (e.g. 1st Bn Guwahati, 8th Bn Ghaziabad) with automated situational awareness reports.

Transparent XAI Causality
Replaces black-box AI scores with physical evidence: rainfall intensity, topsoil saturation deficit, slope gradient, and CWC river stage.

Critical Infrastructure Safeguards
Protects vulnerable populations by routing evacuations to verified high-ground schools, trauma centers, and unflooded arterial road corridors.

Dual-Engine Rescue Triage
Prevents misallocating rescue boats by isolating river breaches from street waterlogging, saving operational emergency response costs by >80%.

Zero-App Multilingual Inclusivity
Transmits emergency text in native mother tongues (Assamese, Odia, Bengali, Nepali, Hindi) with loud acoustic sirens to basic mobile handsets.

DIRECTLY FULFILLS SIH 26192: Generates hyper-local forecasts at village/ward level, providing actionable lead time for evacuation and disaster preparedness.

RESEARCH AND REFERENCES

Evidence supporting the individual building blocks and the integration gap

Central Water Commission (CWC, 2024)
National Hydrology Project (NHP) Standards for river stage gauging, warning/danger limits, and reservoir spillway operations.
https://cwc.gov.in/nhp

ITU-T Recommendation X.1303 / OASIS CAP 1.2
Common Alerting Protocol specification adopted by MHA / NDMA for the National Emergency Cell Broadcast System (SACHET).
https://www.itu.int/rec/T-REC-X.1303

Alfieri et al. (2013)
GloFAS: global ensemble streamflow forecasting and continental flood early warning.
https://doi.org/10.5194/hess-17-1161-2013

Tellman et al. (2021), Nature
Satellite mapping of 913 large flood events + population exposure dynamics.
https://doi.org/10.1038/s41586-021-03695-w

Kratzert et al. (2018)
LSTM rainfall?runoff modelling across 241 complex catchments.
https://doi.org/10.5194/hess-22-6005-2018

NDMA Guidelines (MHA, Govt. of India)
National Guidelines on Flash Floods, Cloudbursts, and Landslide Risk Mitigation in Hilly Ecosystems.
https://ndma.gov.in/guidelines

KEY RESEARCH GAP RESOLVED BY OUR FRAMEWORK:
Existing research is fragmented between isolated academic ML models and disconnected post-disaster communication channels. Our framework closes this critical national gap by uniting multi-source catchment telemetry (IMD radar, GloFAS discharge, CWC gauges), dual-engine fluvial/pluvial triage, and MHA-compliant CAP 1.2 multi-language cell broadcast into an end-to-end operational life-safety system.

Operational Status: 100% live streaming telemetry, 200+ CWC verified station benchmarks, offline resiliency, and real-time citizen ground-truth calibration active in working prototype.